

JACK DRACH

Slot Game Designer • Math & Mechanics • Self-Taught Developer

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PROFILE

Independent slot game designer who builds complete games end to end — math model, RTP tuning, feature mechanics, art direction, and the moment-to-moment pacing that makes a spin feel right. I design and validate every game's math in Python with large-scale Monte Carlo simulation, then build the playable product as a commercial-style slot: a real game engine with outcomes resolved server-side, not faked in the browser. My portfolio of original games is live and playable at jackdrach.com. Seeking a math or game design role at a slot studio.

ORIGINAL SLOT GAMES — JACKDRACH.COM

Super Sundae Scatter

5×5 · 20 lines · 96.56% RTP · PixiJS / TypeScript

- Designed and shipped a complete pay-line slot with a free-spins bonus and a growing wild-scoop mechanic; built on a PixiJS v8 / TypeScript / GSAP / Vite stack.
- Validated the math to a 96.56% return through large-scale Monte Carlo simulation, using direct bonus-round sampling to calibrate the bonus's contribution to overall RTP.
- Architected the game server-authoritative — the client renders outcomes but never decides them — mirroring how real-money regulated slots are built, and deployed it as a live, playable demo.

Ze Claw

7×7 cluster-pays · buy-in feature · JavaScript

- Designed an arcade claw-machine slot with cluster-pays wins, a tumbling board, and a player-choice buy-in (sweep the board clean vs. go for the jackpot pre-seed).
- Built and tuned a per-cell multiplier engine and identified the win-count multiplier compounding as the primary RTP driver, then balanced it through iterative simulation.
- Maintained a Python Monte Carlo simulator as the source of truth for the math, reconciling it against live game behavior to keep the model and the game in lockstep.

Jack's Beanstalk Sky Drop

6×6 cluster tumble · ~96.5% RTP · PixiJS / TypeScript

- Designed a 6×6 cluster-pays tumble slot featuring a sticky wild-stalk mechanic and a board-wide multiplier system, validated to a ~96.5% target return.

Gingerbread's Grand Gables

6×6 all-ways · dual-bonus design · PixiJS / TypeScript

- Designed a festive all-ways slot with two distinct tuned bonus rounds (a raining-symbols feature and a build-up feature), each balanced to its own target RTP.

DESIGN & TECHNICAL APPROACH

- Math & RTP:** Build Python Monte Carlo harnesses that run the real game logic (not a separate model) across millions of spins to validate RTP, hit frequency, bonus trigger rates, and volatility before a game ships.
- Mechanics design:** Cluster-pays, pay-lines, all-ways, tumbling boards, sticky and growing wilds, multiplier systems, free-spins and buy-in bonus features.
- Engineering:** Self-taught developer using Claude Code as a daily build partner to develop, audit, and ship games in TypeScript and JavaScript on a PixiJS / GSAP / Vite stack, with a server-authoritative outcome architecture.
- Production:** Own the full pipeline — concept, math, art direction (AI-assisted asset generation), build, validation, and deployment of a live portfolio site via Git-based CI/CD.

SKILLS

- Game Design:** Slot math modeling, RTP & volatility tuning, payable & weight design, bonus/feature design, player psychology & pacing
- Simulation & Data:** Python, Monte Carlo simulation, statistical validation, regression analysis (R), SQL
- Development:** TypeScript, JavaScript, PixiJS, GSAP, Vite, HTML/CSS, Git, Claude Code, Netlify

EDUCATION

The Pennsylvania State University

B.S., Cyber Security Analytics and Operations

University Park, PA · Aug 2025